Journal of Forest Research

ISSN: 1341-6979 (Print) 1610-7403 (Online) Journal homepage: https://www.tandfonline.com/loi/tjfr20

Impact of the change in raw material supply on enterprise strategies of the Japanese plywood industry

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JOURNAL OF FOREST RESEARCH
2018, VOL. 23, NO. 6, 325–335
https://doi.org/10.1080/13416979.2018.1534048

ORIGINAL ARTICLE

Impact of the change in raw material supply on enterprise strategies of the Japanese plywood industry

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ABSTRACT

The Japanese plywood industry has been profoundly affected by external factors such as raw material supply from overseas, and has increased the utilization rate of domestic timber in recent years. To analyze an industry influenced by external factors, it is useful to employ a theory of competitive factors to examine the industrial structure and the impact on enterprises. In this article, we analyze the correspondence of each enterprise in the Japanese plywood industry for changes in raw material supply and then discuss the predicted future development. The results describe the three major strategies: 1) Differentiation strategy – production with additional processed plywood, continued utilization of tropical wood, and utilizing only domestic timbers; 2) Focus strategy – obtaining certification and locating mills in low-competition regions; and 3) Diversification strategy – product diversification and export. Hereafter, it is predicted that there will be a decline in domestic demand, and consequently, an increase in enterprises that adopt the diversification strategy.

ARTICLE HISTORY
Received 13 March 2018
Accepted 5 October 2018

KEYWORDS
Plywood industry; enterprise strategy; softwood plywood; Japan

Introduction

The self-sufficiency rate of Japanese domestic timber utilization increased by 34.8% in 2016. One of the factors responsible is the drastic increase in the self-sufficiency rate of domestic timber in the plywood industry post 2008, which was due to the gradual increase in domestic timber consumption from the early 2000s and the drastic decrease in Russian timber supply in 2008. Approximately 50% of the domestic plywood demand was supplied by domestic mills in 2016 (JPMA 2016; Japan Forest Products Journal, 2017b February 2). The rate of domestic timber in the total amount of raw material that was utilized in domestic mills became approximately 80% (3.36 million m^3 out of 4.22 million m^3) (MAFF 2016). More than 80% of the domestically produced plywood was structural plywood, therefore, a large part of domestic timber utilization for plywood was for structural plywood. Conversely, the demand for concrete-form plywood was estimated to be approximately 0.7–0.8 million m^3. Of this, more than 90% was imported from overseas and most of the products contained tropical timber from domestic mills; therefore, the domestic timber rate in concrete-form plywood was significantly low (Kawakita 2015; Shibusawa and Kawakita 2016). To further increase the domestic timber utilization rate, its use for non-structural plywood, such as concrete-form plywood and baseboard plywood, for floors was set out (Shibusawa et al. 2014).

In the Japanese plywood industry, the species used for raw material and their origins has changed over time. It began with lauan (*Shorea* spp.) from tropical countries, to larch (*Larix gmelinii*) from Russia, to domestic timber (Tachibana 2000b; Shimase 2007). These changes were due to various factors such as forest resource depletion, export restrictions on logs in Indonesia, the development of technology for softwood plywood, a new distribution and processing program (*Shin Ryuutu Kakou System*),[1] and a significant increase in tariffs on Russian logs. Tachibana (2000b), Shimase (2007), and Tada (2012) clarified the relationship between these changes and factors. Furthermore, the Japanese policy target for increasing the self-sufficiency rate of timber was set in 2009. Changes to the environment surrounding the Japanese plywood industry strongly influenced the strategy of each enterprise.

Porter (1980) has made significant contributions regarding factors contributing to the strategy adopted by enterprises. Five competitive factors were presented: Competitive rivalry (Force 1), the threat of new entry (Force 2), the threat of substitution (Force 3), supplier power (Force 4), and buyer power (Force 5). To deal with these five forces, either of the following three generic strategies can be employed: a) Overall cost leadership strategy, which implies that an enterprise survives competition by producing lower cost products based on effective cost management such as generating a scale economy; b) differentiation strategy, which implies that an enterprise produces and sells products that are clearly distinguishable from other enterprises in the industry; and c) focus strategy, which implies that an enterprise develops an overall cost leadership strategy or differentiation strategy for a specific area and/or target and account. The criticism on this is that the causal relationships between competitive factors and enterprise strategy cannot be assessed statistically, and that this categorization does not readily apply across all industries. Competitive factors are dependent on the enterprises' internal characteristics and performances; therefore, the same hypothesis should not be applied to all enterprises (Barney 1991). However, to analyze an industry that is influenced by external factors, it is useful to employ the theory of competitive factors for an in-depth analysis of the industrial structure and for discussing the impact on enterprises (David et al. 1997). It can be argued that the Japanese plywood industry is highly influenced by external factors

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such as raw material supply from overseas. Until now, no analysis has been conducted on the enterprise strategy concerning the Japanese plywood industry. In this article, we aim to clarify the processes involved with increasing the utilization rate of domestic timber as well as the influences, and aim to predict the trend in the industry. For this purpose, we analyze the correspondence between changes in the raw material supply and products and the enterprise strategy for each enterprise in the Japanese plywood industry. We chronologically present the changes of raw material species and its origin, the factors contributing to these changes, and the trend of the plywood industry. We employ the theory of competitive factors by Porter (1980) to analyze and then discuss the strategies adopted to deal with the competition among enterprises.

Methods

We selected 17 large-scale plywood enterprises[2] by gathering information from their webpages and documents of enterprises and certifications regarding their management. The information collected includes: the starting year of plywood production, the year during the addition of mills and the number of mills added, the year of changing the species used for raw material, and the origin of the material, the year of changing plywood products,[3] the condition for producing non-plywood products, the year of attaining certification, and the year of starting wood biomass power generation. Out of 17 enterprises, there were 15 that belonged to enterprise groups (Table 1). There were a total of four enterprise groups. The largest enterprise group consisted of eight individual enterprises. The other three groups consisted of either two or three enterprises.

The aggregated information was verified for its relationship with major events or policies that have influenced the plywood industry. Next, we analyzed and discussed this information with reference to the theory of competitive factors (Porter 1980) based on each enterprise's business policy.

Regarding enterprise strategies, when enterprises are placed in a competitive industrial environment, these enterprises employ an overall cost leadership strategy, differentiation strategy, and/or focus strategy (Porter 1980). Among the three strategies, discussion on the overall cost leadership strategy in the plywood industry requires each enterprise's specific information regarding cost minimization such as the purchase price of raw material, the recovery rate, and the transition of production volume. However, such information was not readily available on the webpages or in the documents. Therefore, we mainly focused on the differentiation strategy and the focus strategy. Furthermore, we tallied the keywords of the enterprises' webpages regarding the image of enterprises and their products to categorize appeal points as a means of differentiation.

Kotler and Keller (2012) noted that industry trends can be outlined based on the perspective of the product life cycle (PLC), and could be categorized into four stages (introduction stage, growth stage, maturity stage, and decline stage). Enterprises should modify strategies based on these stages of change. We also employed the perspective of the PLC (Kotler and Keller 2012) as a benchmark to predict the trend of industry.

All the information listed above was collected from October 2016 to June 2017.

Table 1. Belonging group and located area of targeted enterprises in this article.

	Group				
Area	A	B	C	D	No grouped
Hokkaido					1
Tohoku	5	1			
Chubu	1	1	2		1 (Chubu + Kinki)
Chugoku	1			3	
Kyusyu	1				
Total	8	2	2	3	2

Source: Webpages of targeted enterprises in this article

Results

Historical trend of Japanese plywood

After World War II, the domestic plywood industry had consistently used tropical timber for their production until 1985, even though the timber supply countries had changed from the Philippines to Malaysia and Indonesia (Figure 1) (Tachibana 2000a). However, the plywood industry in Indonesia was developed due to the restriction on log exports in 1985, and thus, a large amount of plywood products was then exported to Japan (Lönnstedt 1996; Araya 1998). The rate of imported plywood exceeded domestic plywood production in 1996. After 1996, the amount of domestic production continued to decrease. However, a new distribution and processing program implemented from 2004 to 2006 caused the utilization of timber from forest thinning that was felled but lain aside in forest before the implementation of the program, which increased the domestic timber utilization and domestic production (Tada 2012). Even though the amount of production had temporarily decreased in the latter half of the 2000s due to increased tariffs on Russian logs in 2007 and 2008 and the Great Recession in 2008 (i.e. the collapse of the Lehman Brothers), the consumption of domestic softwood for plywood continued to increase. In contrast, the amount of imported plywood gradually decreased, and 52.5% of the plywood demand in Japan was supplied by domestic mills (Magazine for Building Materials 2017).

Figure 2 illustrates the amount of timber consumption for domestic plywood production. In the 1970s, 95% of timber consumed in the domestic plywood industry was imported from Southeast Asia (Murashima 1988). In the 1980s, 80% of the tropical timber imported into Japan was for the plywood industry (Okano and Iwai 1990; Lönnstedt 1996; Sasse 1998). Large amounts of tropical timber were consumed. However, after restrictions on log exports in Indonesia were applied in 1985, the amount of domestic plywood products that utilized tropical timber decreased (Tachibana 2000b). Due to the technological advances in softwood plywood, softwood, including Russian timber, was able to compensate for the decline in tropical timber imports (Lönnstedt 1996; JAWIC 1998). However, the import of Russian timber declined significantly due to a large increase in log tariffs in 2007 and 2008 (Yamane 2013). From 2004, that is, the year of implementation of the new distribution and processing program, enterprises that utilized domestic timber increased (Endo 2015). Consequently, the utilization rate of domestic timber reached 80%.

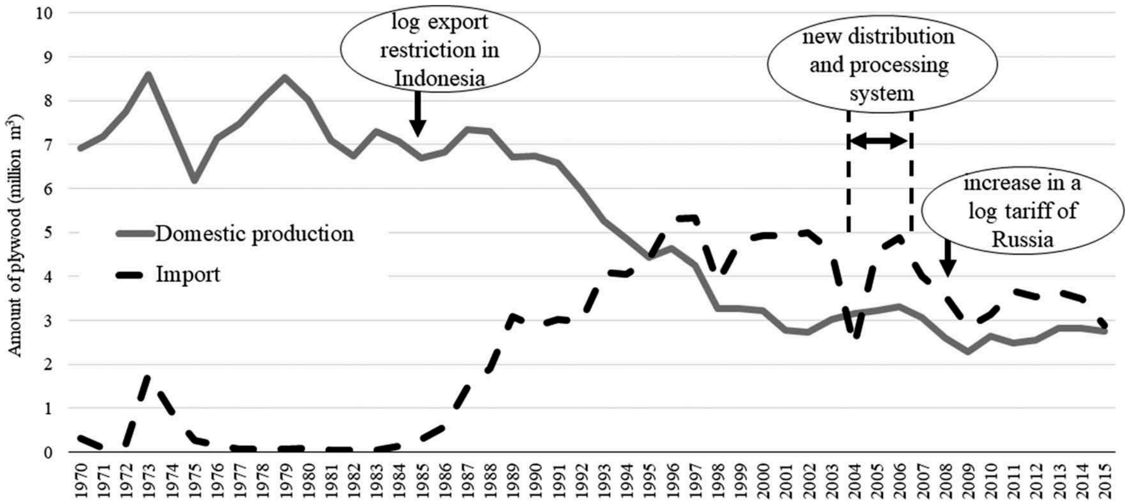

Figure 1. Changes in the amount of domestically produced and imported plywood.
Source: JPMA (2016).

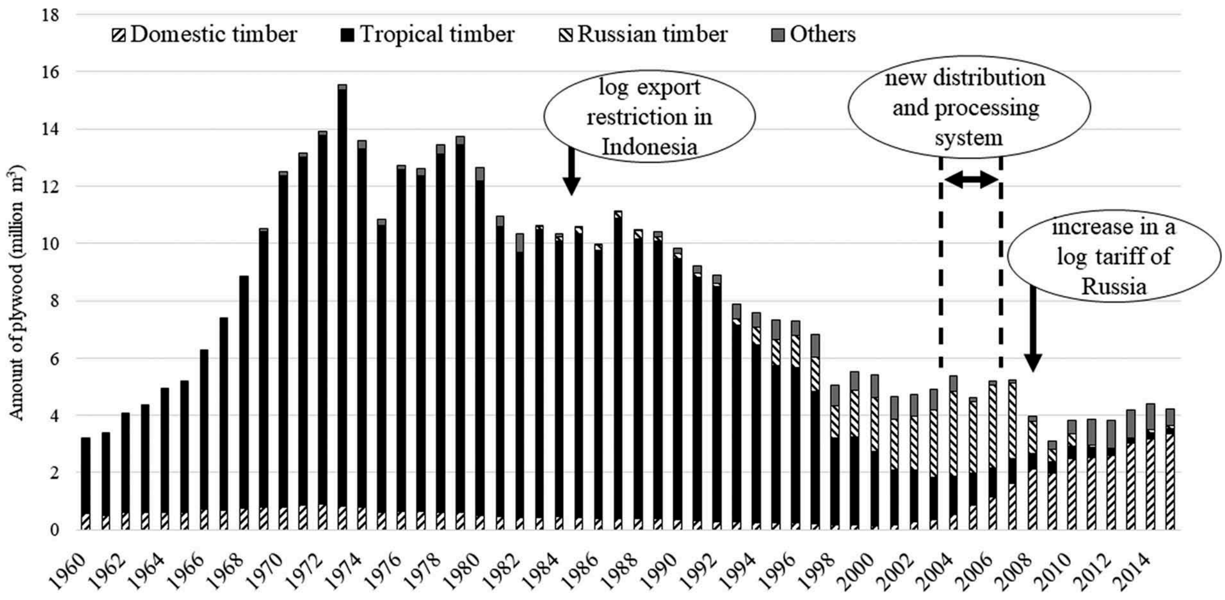

Figure 2. Changes in the amount of timber utilization for domestically produced plywood.
Source: JPMA: Gouhan toukei [Statistics of plywood] (2016) and MAFF (2016).

With an increase in domestic plywood production, the amount of production per mill increased continuously and the production scale had more than doubled for mills that produce plywood for general use (Figure 3).[4] The number of domestic mills that produced plywood for general use decreased from 82 in 1998 to 34 in 2015. This indicates that small-scale mills could not compete with the mass production carried out by large-scale mills.

Trends of domestic plywood enterprise

It was suggested that the strategies of enterprises were influenced by events such as the restrictions on log exports in Indonesia in 1985, the introduction of a new distribution and processing program from 2004 to 2006, and the increase in Russian log tariffs in 2007 and 2008. These aspects also influenced the trends in the

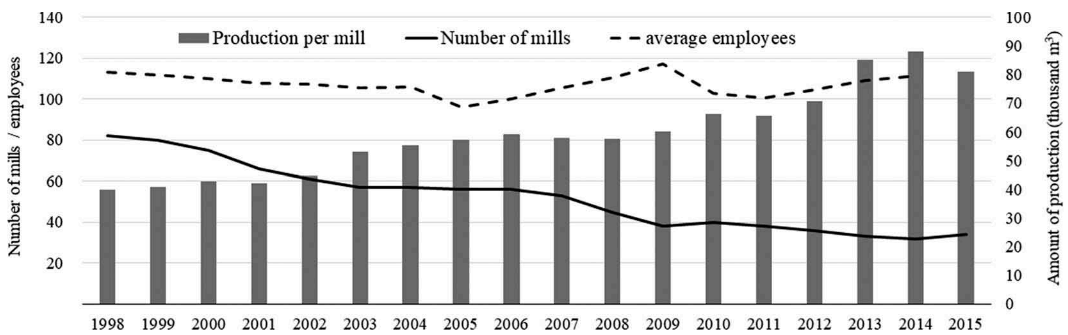

Figure 3. Changes in the scale of domestic plywood mills.
Source: MAFF: Mokuzai jukyuuhyou [Timber balance sheet] (2016).

plywood industry such as the changes in the source of raw material, from tropical timber to Russian timber, and then to domestic timber. Here, the strategies adopted by enterprises were shown along with the trends of the plywood industry.

Adding mills and new entry of enterprises

To respond to the growing demand for domestic plywood, 10 mills were established and three enterprises newly entered the industry in the 1970s to produce plywood utilizing tropical timber that continued to increase import (Figure 4). In the 1990s, to respond to the change in softwood plywood due to the restrictions on log exports in Indonesia in 1985, six mills were added. The Russian log tariff increase in 2007 and 2008 led to a decrease in the import of Russian timber. However, both Russian timber and domestic timber were of softwood, and therefore, the necessity for mechanical equipment change, starting operations, or adding mills was relatively low. In recent years, there were two cases of new entry of enterprises utilizing domestic timber and two cases of adding mills. These two enterprises belonged to the largest enterprise group and produced structural plywood only using domestic timber. Furthermore, they were not located along the coastline, which would be ideal for access to imported raw materials, like the already existing mills. Rather, these were located inland, close to the local timber.

Changes in material tree species

Before 1975, no enterprise changed or added the material tree species (Figure 5). In the 1980s and 1990s, four enterprises changed the material tree species from tropical timber to softwood timber due to the restrictions on log export in Indonesia in1985. In the early 2000s, during years of gradual increase of domestic timber consumption and steady rise of Russian timber imports, three enterprises changed or added tree species used for source material. At that time, it was considered that almost all enterprises started or attempted to utilize domestic softwood timber. Therefore, the tree species utilized for material did not change after the increase in Russian log tariffs in 2007 and 2008.

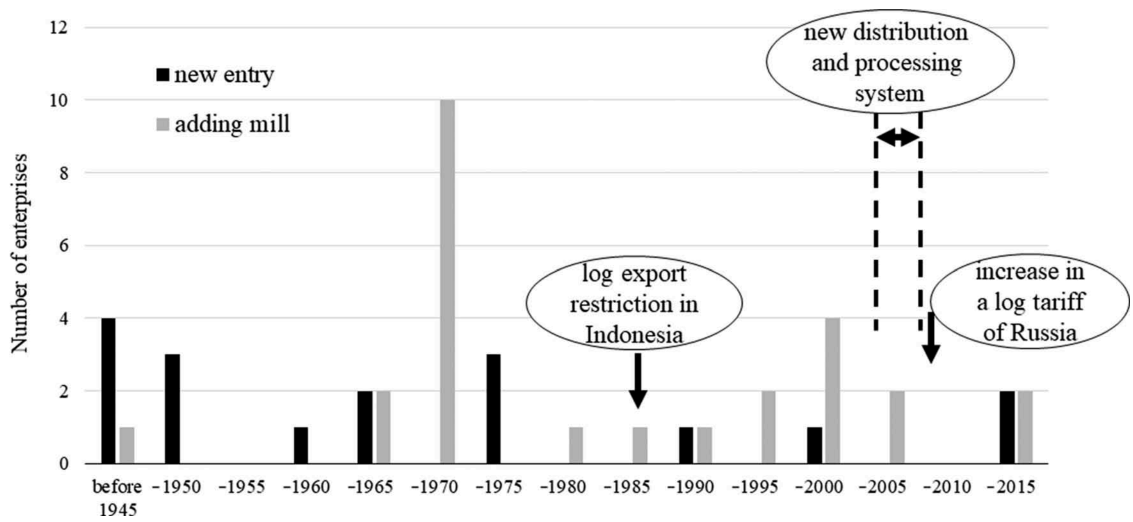

Figure 4. The number of domestic plywood enterprises starting operation and adding mills.
Source: Webpages of targeted enterprises in this article

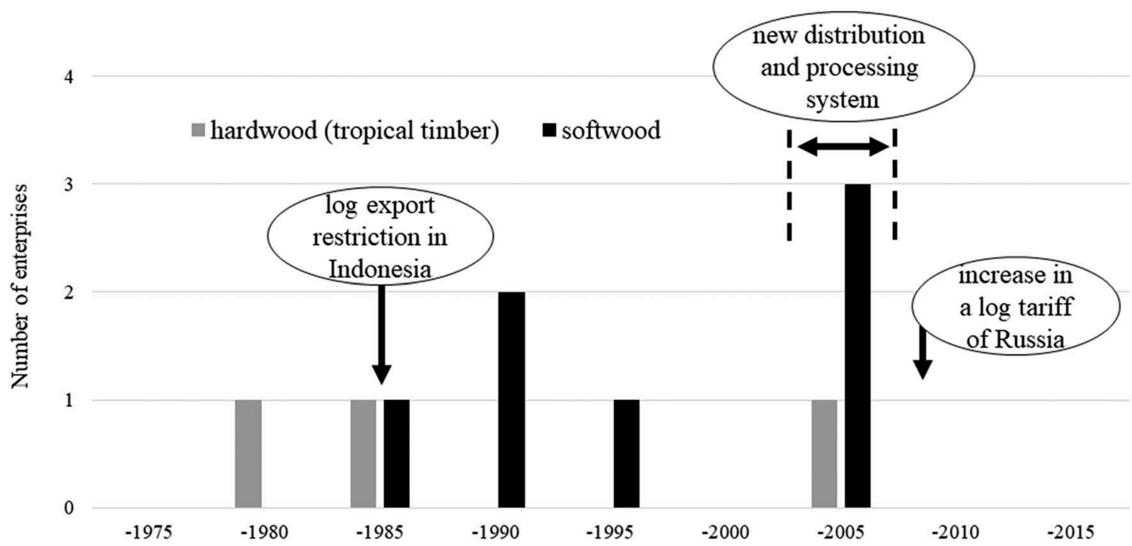

Figure 5. The number of domestic plywood enterprises changing or adding the material tree species.
Source: webpages of targeted enterprises in this article.

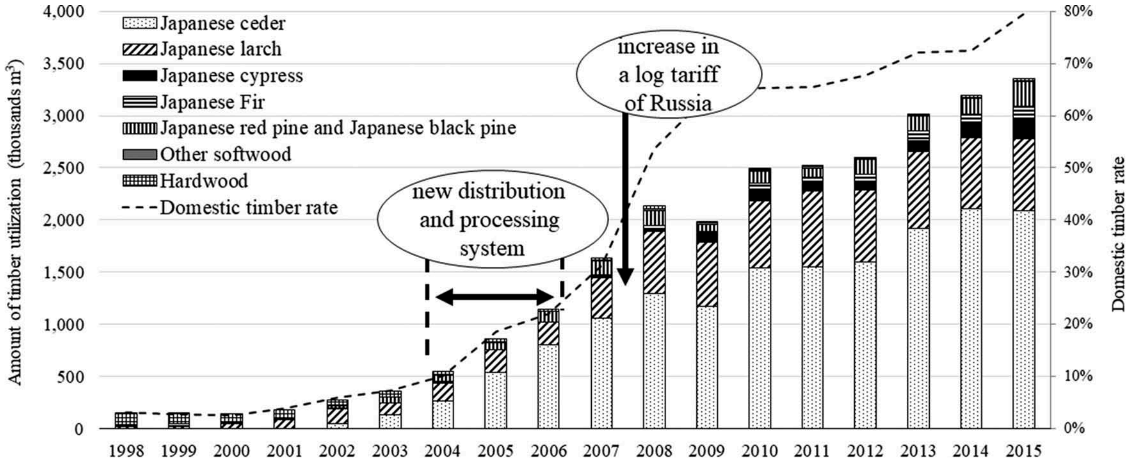

Figure 6. Changes in the amount of domestic timber utilization in domestic plywood mills.
Source: MAFF: Mokuzai jukyuuhyou [Timber balance sheet] (2016).

The records of the Ministry of Agriculture, Forestry and Fisheries in Japan exhibited that the amount of domestic timber consumed at domestic plywood mills began increasing since the implementation of the new distribution and processing program from 2004 to 2006 (Figure 6) (MAFF: Mokuzai jukyuuhyou [Timber balance sheet] 2016). Upon the decrease in Russian timber supply, especially of larch, due to the significant log tariff increase in 2007 and 2008, the consumption of Japanese larch (mainly *Larix kaempferi*) increased to approximately the same amount as in 2015. The increase in the consumption of domestic timber was achieved by increasing the utilization of Japanese cedar (*Cryptomeria japonica*). In 2015, 2,087,000 m^3 of Japanese cedar was consumed, making up approximately 62% of the domestic timber for plywood production.

In 2015, the utilization rate of domestic timber in domestic plywood production became 80% (Figure 6); however, at same time several enterprises utilized timber from the United States, Russia, tropical countries, and New Zealand (Table 2). Domestic timber that was utilized the most was Japanese cedar (utilized by 15 enterprises). Larch was also utilized by 14 enterprises, regardless of its origin. To attain the appropriate strength for plywood, Douglas fir (*Pseudotsuga menziesii*), Russian larch, Japanese larch, and Japanese cypress (*Chamaecyparis obtusa*) were utilized for the face and back layer of the plywood. Therefore, there was a high demand for these species. There were two enterprises that were still producing plywood using tropical timber, especially lauan.

Table 2. Number of enterprises utilizing each material species.

Origin of timber	Name of tree species (scientific name)	Number of enterprises
Japan	Japanese cedar (*Cryptomeria japonica*)	15
	Japanese larch (mainly *Larix kaempferi*)	10
	Japanese cypress (*Chamaecyparis obtusa*)	9
	Japanese fir (*Abies sachalinensis*)	3
	Japanese black pine (*Pinus thunbergii*)	1
	Japanese red pine (*Pinus densiflora*)	1
USA	Mainly Douglas fir (*Pseudotsuga menziesii*)	8
Russia	Mainly larch (*Larix gmelinii*)	5
Tropical countries	Mainly lauan (*Shorea* spp., etc.)	5
New Zealand	Radiate pine (*Pinus radiate*)	2
No data	Larch (*Larix* spp.)	3

Source: Webpages of targeted enterprises in this article and Japan Forest Products Journal (2016November 3)

Products

In the latter half of the 1960s, seven enterprises began producing diversified products, such as concrete-form plywood using tropical timber, or thin plywood to respond to the diversified demand from consumers (Figure 7). A major change from tropical hardwood products to softwood products could be noted in 1985 due to the restrictions on log exports in Indonesia in 1985. However, post the restrictions, there were no further major changes in products because the domestic plywood industry had entered the growth period in producing structural plywood.

Table 3 indicates that all enterprises produced structural plywood and 13 out of all enterprises produced *Nedanon*.[5] *Nedanon* is a type of structural plywood that mainly uses domestic timber. It is much more resistant to earthquakes than conventional products (Tokyo Plywood Manufacturers' Association and Tohoku Plywood Manufacturers' Association 2016). In 1999, *Nedanon* was formally approved by the Japan Plywood Manufacturers' Association. It played an extraordinary role where it increased the shipping volume of softwood plywood by 17.5% compared to the previous year (Miyazaki 2010). Concrete-form plywood was produced by 11 enterprises. Until recently, concrete-form plywood was domestically produced using tropical timber or imported plywood products from tropical countries. However, nine out of eleven enterprises produced concrete-form plywood using domestic softwood timber. Six enterprises produced value-added plywood with additional treatments such as preservation and ant-proof, water resistant/repellent, flame-resistant, and formaldehyde-free. Only a few enterprises from the same enterprise group or region produced the plywood with additional processing or treatment. Three enterprises produced structural plywood only and did not produce plywood with additional processing or treatments. Two of them had recently entered into the industry after 2011.

Some enterprises started to diversify by producing non-plywood products. Seven enterprises produced laminated veneer lumber (LVL), four enterprises produced medium-density fiberboards (MDF), three enterprises produced particle boards (PB), and one enterprise produced cross-laminated timber (CLT). Two enterprises produced three of these products. These enterprises produced not only LVL, which has a production process similar to that of plywood, but also MDF as a baseboard for floors. These enterprises conducted integrated production within one enterprise.

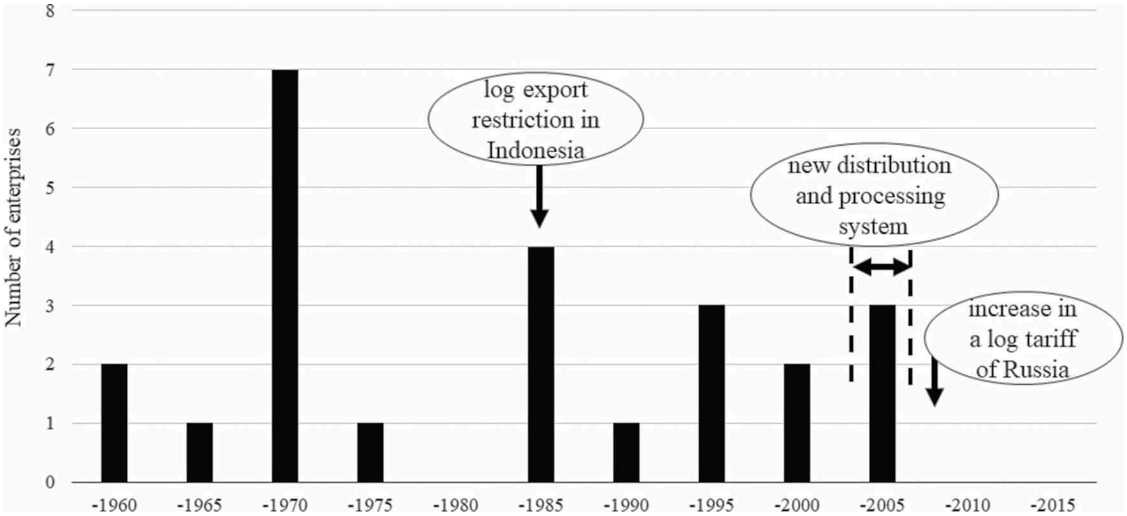

Figure 7. The number of domestic plywood enterprises changing or adding products.
Source: webpages of targeted enterprises in this article.

Table 3. Number of enterprises by product category.

Product	Number of enterprises
Structural plywood	17
(*Nedanon*)	(13)
Concrete-form plywood	11
Baseboard plywood for floor	6
Additional processing/treatment	
Preservation/ant-proof	5
Water-resistant/repellent	3
Flame-resistant	1
Formaldehyde-free	1

Source: Webpages of targeted enterprises in this article

Quality of products

Attaining quality assurance standards and environmental certification may be effective in targeting specific customers, and for being able to charge a price premium. Attaining these is meaningful for customers who wish to buy certified products as a focus strategy or for enterprises that want to clearly distinguish themselves from their competitors' products as a differentiation strategy.

All enterprises had attained the Japan Agricultural Standard (JAS[6]) enacted in 2003 regarding quality management. Eleven enterprises attained the certification standard ISO 9001 by the International Organization for Standardization, which was established in 1987 (Figure 8). As mentioned above, producing *Nedanon* gives products the appeal of high earthquake-resistance products.

Environment consciousness

Environmental certification and forest certification also have the potential of attracting specific customers. Regarding environmental certification, six enterprises attained ISO14001, which was established in 1996 (Figure 9). Attaining forest certification is one of the methods for proving legality in abiding by the Clean Wood Act (2017)[7]; therefore, enterprises having a forest certification receive significant attention, and thus, it is being extensively attained by enterprises. In 2016, regarding the certification of products in the processing and circulation steps, namely the Chain of Custody (CoC) certification, 11 enterprises attained certification by the Forest Stewardship Council (FSC), which was established in 1993. Seven enterprises attained certification by the Programme for the Endorsement of Forest Certification (PEFC), which was established in 1999; and eight enterprises attained certification by the Sustainable Green Ecosystem Council (SGEC), which was established in 2003 (FSC 2016; PEFC 2016; and SGEC 2016). The number of enterprises that attained at least one of these three certifications was 14 (82%).

There were five enterprises that started or had planned to generate electric power using wood biomass. Many enterprises generally utilized mill residue for boiler operations. Enterprises that generated electric power from wood biomass have increased after the mid-2000s to create an environmentally consciousness image as well as to reduce the electricity costs.

Keywords in business policy

To understand enterprises' appeal points, keywords for the image of enterprises and products on the enterprises' webpages were collated (Table 4). More than half of all the enterprises used the keywords "environment consciousness" and "domestic timber." This can be explained by the fact that the Japanese plywood industry had taken the brunt of the criticism regarding the tropical forest problem (Dauvergne 1997). This appealed to environmentally conscious efforts and the utilization of domestic timber for the vitalization of Japanese forests. These enterprises tried to put these efforts into practice by attaining CoC certification (Figure 9) and generating electric power using wood biomass, as well as by indicating the domestic timber mark.[8] Fifteen enterprises (approximately 88%) registered and indicated the domestic timber mark. Regarding the keywords "technological capability" and "high quality," attaining JAS and ISO 9001 or producing *Nedanon* assures a specific level of quality (Figure 8). Many keywords aimed to build credibility with the reality of each enterprise's activities. The two enterprises that used the keyword "utilization of local timber" were located in the southernmost (Kyushu) and northernmost (Hokkaido) regions of Japan. These enterprises produced plywood with the maximum scale in each region using local timber.

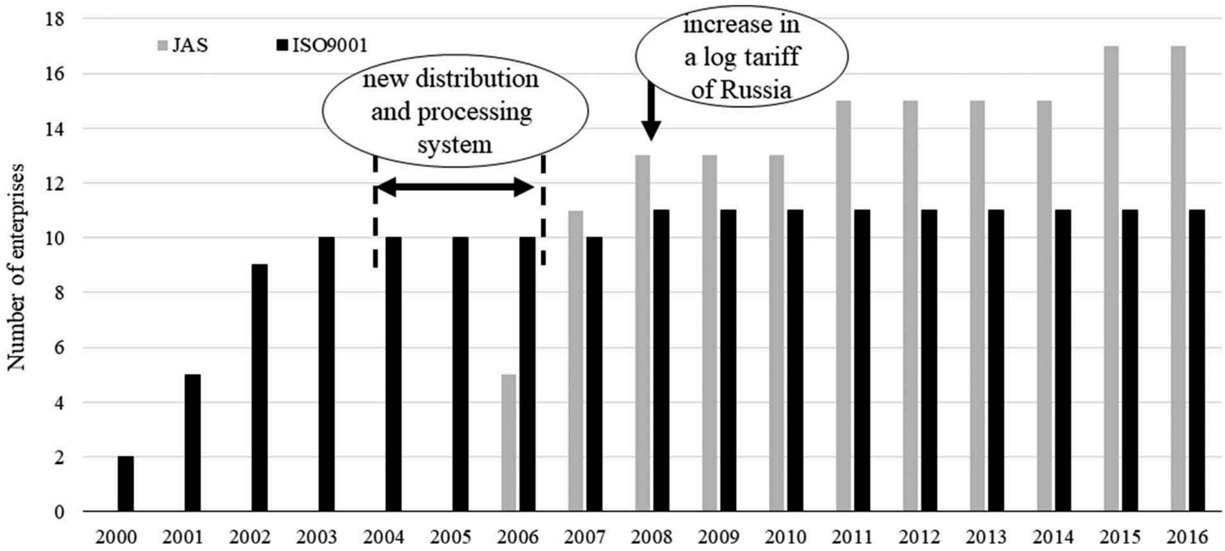

Figure 8. The cumulative number of domestic plywood enterprises obtaining quality certification.
Source: webpages of targeted enterprises in this article, JAB (2016) and JPIC (2016).

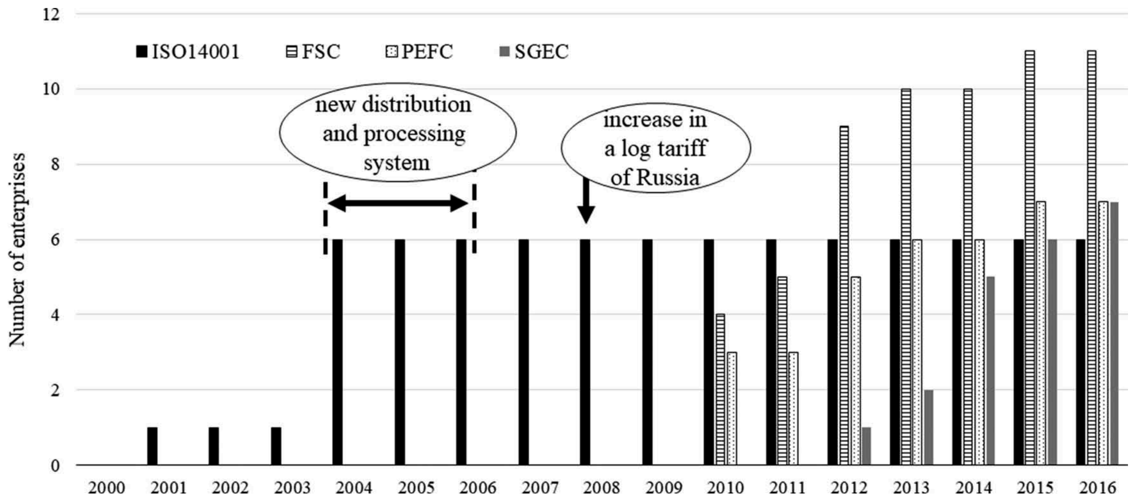

Figure 9. The cumulative number of domestic plywood enterprises obtaining environment certification.
Source: webpages targeted enterprises in this article, JAB (2016), FSC (2016), PEFC (2016), and SGEC (2016).

Table 4. The number of enterprises posting keyword on webpages.

Keyword	Number of enterprises
Environment consciousness	10
Domestic timber	10
Technological capability	6
Utilization of wood resources	5
Stable supply	3
High quality	2
Utilization of local timber	2

Source: Webpages of targeted enterprises in this article

Discussion

In this article, statistic information regarding plywood production was indicated on the basis of events concerning the Japanese plywood industry. It was reaffirmed that the major factors that influenced this industry were the restrictions on log exports in Indonesia in 1985, the implementation of the new distribution and processing program from 2004 to 2006, and the increase in Russian log tariffs in 2007 and 2008.

Furthermore, the current state of the plywood industry was discussed along with the five factors presented by Porter (1980). Regarding competitive rivalry (Force 1), the plywood industry had the following characteristics: high inventory costs, high price of mechanical equipment, and a high exit barrier. Therefore, the industry had a structure that strengthens an adversarial relationship among enterprises in the business. Furthermore, the locations of enterprises are concentrated in northeastern (Tohoku), central (Chubu), and western (Chugoku) parts of the Japanese main island. Decreased plywood demand was estimated as a result of a decline in new housing, which was further associated with a decrease in birthrate and population; therefore, competition for shipment in each area will occur (Table 1). Hence, inter-group or inter-enterprise competition is expected to be fierce. At this time, the plywood production utilizing imported tropical timber or imported plywood products has decreased because the amount of tropical timber supply has decreased, whereas the amount of domestic timber supply has increased (Figure 1).

Regarding the threat of new entry (Force 2), there were only few new entries because the plywood industry was an apparatus and capital-intensive one. Only under the biggest enterprise group could new enterprises enter the industry (Figure 4).

Regarding the threat of substitution (Force 3), we should discuss from three aspects; raw materials, place of production, and non-plywood products. Threat of substitution in raw materials were among tropical timber, Russian timber, and domestic timber. Before the 2000s, plywood supplied in Japan was mainly produced from tropical timber regardless of the place of production. From the middle of 2000s, Russian timber consumption gradually increased until 2007, then drastically dropped in 2008 due to the Russian log tariff increase. On the one hand, from the early 2000s, domestic timber consumption gradually increased, then became the main raw materials from 2008. Until now, the position regarding raw materials between the main and substitute had changed twice.

Threat of substitution in place of production were between Japan and tropical countries. Imported tropical plywood began increasing from 1985 due to the development of the Indonesian plywood industry resulting from the restriction on log exports. The main plywood product was reversed from domestically produced plywood to imported plywood in 1995. However, in 2016, the main plywood product was reversed again to domestically produced plywood because of the gradual decrease in imported plywood from the latter half of the 2000s (Figure 1).

Non-plywood products are necessary to regard such as steel concrete-form and precast concrete[9] for concrete-form plywood and oriented strand board (OSB) for structural plywood as substitutions, even though the market share for these are still small and the prices are high.

In the plywood industry, "supplier" indicates "timber producer," and the bargaining power of timber producers (Force 4) became stronger as Japanese larch timber was in high demand because of its strength (Table 2). It will be important to observe how a stable timber supply can be ensured when the utilization rate of domestic timber becomes higher. It is expected that the bargaining power of timber producers will become even stronger because the production of timber will ensure accelerated competition both in the plywood industry and other businesses such as those generating electric power from wood biomass.

Regarding buyer power (Force 5), it is expected that buyers will sort out the sellers because builders have the possibility to purchase only certified products for the preparation of Tokyo Olympic and Paralympic Games in 2020.[10] What relates to this movement that there were many enterprises that attained ISO or CoC certifications for increasing their bargaining power over buyers (Figures 8 and 9).

Next, the correspondence of each enterprise was analyzed. A differentiation strategy served as a competitive one for enterprises in the plywood industry such as for producing additionally processed plywood (Table 3), the continuing utilization of tropical wood (Table 2), and utilizing only domestic timber. Three out of six enterprises were producing additionally processed plywood. Two enterprises that were utilizing only domestic timber belonged to the largest enterprise group (Table 5). It can be understood that these enterprises had adopted a differentiation strategy even though they belonged to the largest enterprise group. Producing plywood utilizing only domestic timber equals to a differentiation strategy at this time. However, the utilization rate of domestic timber in the plywood industry continues to rise. Therefore, it is expected that domestic timber utilization will eventually become the new normal. Two enterprises out of the remaining three were continuously producing additionally processed plywood while mainly utilizing tropical timber. Therefore, it can be said that these enterprises adopted an extreme differentiation strategy.

Table 5. Summary of specific materials or production regarding strategies by belonging group.

Products and materials	Group A	B	C	D	No grouped	Total
Products						
Plywood with any additional processing/treatment	3	1	2			6
Only structural plywood without additional processing/treatment	3					3
Unclassified	2	1	0	3	2	8
Utilized materials						
Only domestic timber	2					2
Mainly tropical timber			2			2
Unclassified	6	2	0	3	2	13

Source: Webpages of targeted enterprises in this article

Attaining certification is part of the focus strategy of producing and selling intensively for a specific market and/or target account. Difference in the quality of products between the enterprises was not much as all enterprises attained JAS. In addition, 13 enterprises (approximately 76%) produced the unified standardized product, *Nedanon*. These results indicate that attaining quality certification is not for reaching a specific market and account, but is as normal to indicate a certain quality of products (Table 3 and Figure 8). Environmental certification remains one of the focus strategies under existing conditions (Figure 9). However, more than 80% of enterprises attained at least one of any three main schemes of certification. These certifications are being considered as one of the methods for proving legal timbers use in abiding to the Clean Wood Act (2017), and builders have the possibility of purchasing and using only certified products for the construction of buildings for the Tokyo Olympic and Paralympic Games. These indicate that environmental certification will also be considered as the new normal in the future. Enterprises belonging to the largest enterprise group were comparatively ubiquitous (Table 1). Therefore, there was a possibility that Group A as the largest enterprise group also adopted a strategy of optimal location of mills. This strategy regarding geographical location is also one of the focus strategies. Three enterprises produced only structural plywood and did not provide additional processing or treatment focus on one product (Table 5). It is evident that these enterprises also adopted the focus strategy.

There were some enterprises that offered diversifying products in conjunction with a differentiation strategy and a focus strategy. Diversification of products is the strategy that prolongs the life of enterprises and enterprise groups. Of the two enterprises that offered diversifying products, one was the only large-scale enterprise in the region and the other was an enterprise belonging to the largest enterprise group. Even though diversification has its own risks, all products use the same distribution channel as building materials. Therefore, this is an effective strategy. The diversifying products are wood boards (i.e. LVL, MDF, PB, and CLT). Furthermore, export indicates the diversification of the market. Thus, export is also one of the diversification strategies. With regard to the expected continuing population decrease and decline of the domestic demand, the enterprise belonging to the largest enterprise group began developing the overseas markets, while simultaneously enlarging their market share in the domestic market and in keeping with the mission of the largest enterprise group, namely expansion of the total market.

Regarding an overall cost leadership strategy, the information is limited; however, it is specified below. Group A, the largest enterprise group (Table 1), was considered to have been adopting an overall cost leadership strategy based on Porter's theory. Therefore, we summarized the strategy of Group A as a case. Group A offered the diversification of products and exported to expand the total market. Further, Group A attempted to increase the share by adding mills and new entry of enterprises into the plywood industry, which was even an apparatus and capital-intensive. On the one hand, these enterprises which newly entered also adopted a differentiation strategy by utilizing only domestic timber (Table 5). On the other hand, these enterprises adopted a focus strategy by producing only structural plywood without providing additional processing or treatment (Table 5). Another enterprise belonging to Group A also adopted a focus strategy by allocating mills in consideration of optimal locations. Thus, Group A adopted not only an overall cost leadership strategy but also the other two strategies. It made efforts for the expansion of the total market rather than cost reduction as the largest enterprise group.

In the situation where promoting the utilization of domestic timber as forestry policy, many enterprises used the keyword "domestic timber" on their webpages. The keyword "environmental consciousness" was also used by many enterprises considering that the plywood industry had received criticisms regarding the tropical forest problem. Five enterprises started or have plans for generating electric power using wood biomass as actual activity for environmental consciousness. It could be understood that "environmental consciousness" and "domestic timber" have become important topics for the Japanese plywood industry (Table 4).

Conclusion

The effectiveness of adopting a differentiation strategy, which utilizes only domestic timber is expected to diminish, because utilizing domestic timber will become the new normal. In contrast, the differentiation strategy of continuing tropical timber utilization has the possibility of further increasing differentiation because of the utilization rate increase of domestic timber (Table 5). Attaining certification, such as CoC certification, as part of a focus strategy will be the normal for achieving mutual recognition between PEFC and SGEC in 2016, and for proving legal timber in accordance with the Clean Wood Act (2017), as well as due to the rising demand of certified timber for the Tokyo Olympic and Paralympic Games in 2020. Regarding the overall cost leadership strategy, the expansion of the total market of plywood by the largest enterprise group will be continued, with product diversification and export of products. By the time utilizing domestic timber and attaining certification have become normal, other enterprises belonging to other groups are also expected to offer diversification of products and to export them. The first-mover advantage of the largest enterprise group will influence the future trends.

The PLC of the Japanese plywood industry has three stages: the growth and maturity stage, where plywood was produced with tropical timber in domestic mills until the mid-1990s; the decline stage, where a huge amount of tropical timber for plywood was imported from the mid-1990s to the mid-2010s; and recently the re-growth stage, where more than half of the domestic demand was met by production from domestic mills. Furthermore, the amount of concrete-form plywood made in Malaysia, which was the largest producer, was reducing. Eleven enterprises (65%) produced concrete-form plywood (Table 3) to recapture the market share, and the largest enterprise group planned to establish a mill for painted concrete-form plywood (Japan Forest Products Journal 2017d March 8). Six enterprises produced baseboard plywood for floors (Table 3). Many enterprises made a strong effort to increase their sales of non-structural plywood, including concrete-form plywood and baseboard plywood for floors (Japan Forest Products Journal 2017c March 4). Therefore, the Japanese plywood industry will continue to grow for a while. After the production of domestic plywood and the utilization rate of domestic timber have reached a new normal level, the industry would have reached a different phase in the maturity stage. If competitive products such as steel concrete-form are able to reach a growth stage, or alternatively, construction methods such as precast concrete will diffuse, the Japanese plywood industry should be concerned about maintaining and enlarging the plywood market to avoid the decline stage. Diversification strategies, such as diversifying products and exports, can be useful countermeasures. Combinations of life cycles between different products under a diversification strategy can promote growth in enterprises. The predicted decrease in housing will begin in 2020 (NRI 2017), which will lead to an increased bargaining power of buyers due to the declining domestic plywood demand. Therefore, it is expected that more enterprises will start adopting an export strategy to avoid the declining amount of plywood supply.

Notes

1. This program was implemented by Forest Agency from 2004 to 2006 for the purpose of straightening the foundation of timber distribution and processing, along with the policy of tree thinning management for utilizing small diameter timber in plywood mills. Furthermore, Tada (2012) pointed out that the introduction of machines such as a rotary lathe was promoted by other subsidies from the country and the prefecture governments that was given prior to a new distribution and processing program. These subsidies had effect to solute technical problems for utilizing domestic timber, especially in plywood enterprises of northeastern part (Tohoku) of Japanese main island, the most concentrated area of plywood enterprises.
2. The accurate amount of plywood production was not released to the public; therefore, we defined large-scale plywood enterprises as those that were estimated to have a production of more than 50,000 m^3. We targeted these enterprises.
3. This implies the difference of usage application for structural plywood, concrete-form plywood, and baseboard plywood for floors. It is defined as a non-change in production (for example, the utilized tree species changed from hardwood to softwood, but the production of structural plywood remained the same).
4. In the statistics of Ministry of Agriculture, Forestry and Fisheries, plywood mills are divided into four types depending on the products: veneer only, plywood for general use only, plywood for general use and special plywood, and special plywood only. Special plywood is plywood that has undergone additional processing and/or treatment, particle board with incrustation of plywood or veneer, and so forth. Here "the mills that produce plywood for general use" means the combined value of two types of mills that plywood for general use only and plywood for general use and special plywood. The mills of targeted enterprises in this article are also included in these two types of mills.

5. *Nedanon* is the mutual name of structural plywood that was registered for the Japanese Agricultural Standard (JAS) by accessed enterprise members of the Tokyo Plywood Manufacturers' Association and Tohoku Plywood Manufacturers' Association.
6. At this moment, JAS was not given the name "certification" but instead was called "standard." Draft revision of the Act for JAS was approved in the Cabinet on 28 February 2017 and the plan is to enact it within a year. In this draft revision, it was mentioned to change the name from "standard" to "certification" (Japan Forest Products Journal 2017a March 3).
7. This act was enforced on 20 May 2017. The official name is "the act concerning circulation and utilization of legally logged timber and timber products" and it is commonly called the "Clean Wood Act." This act aims to promote the circulation and utilization of timber and timber products that are legally logged, respecting the acts of Japan and those of the origin countries of timber. Plywood is included as a timber product (first clause of Article 2). A trader concerning timber and timber products voluntarily applies for registration to be a "registered trader concerned timber and timber products." In regard to this, there are two classes; the first class is a trader who buys timber directly and the second class is a producer of such laminated timber and paper. A plywood producer is included in the first class. If registered, a trader is required to handle all timber legitimately. This act was not yet translated into the English language, so the authors translated it themselves.
8. Institution aiming to disseminate the meaning and importance of domestic timber utilization among Japanese citizens by promoting consumers to choose and utilize domestic timber production, and to contribute to Japanese forest regeneration through appropriate use of the mark indicated by domestic timber product (Office for Domestic Timber Mark Promotion 2017).
9. Construction method that produces concrete member in a dedicated mill and construction on site.
10. For timber utilized in establishing facilities and furniture of such stadium regarding the Tokyo Olympic and Paralympic Games, five procuring conditions were set; "(i) Timber that is harvested through an appropriate procedure with reference to relevant laws, ordinances, etc. of timber-producing countries or territories, (ii) Timber that derives from forests maintained and managed based on mid- to long-term plans or policies, (iii) Timber that is harvested through logging activity that is considerate toward conservation of the ecosystem, (iv) Timber that is harvested through logging activity that is considerate toward the rights of indigenous people and other local residents, and (v) Timber that is harvested by workers protected through appropriate safety measures," and "Timber with FSC, PEFC, or SGEC certification is accepted in principle as one that satisfies conditions (i) to (v)" (The Tokyo Organising Committee of the Olympic and Paralympic Games 2017).

Acknowledgments

This study was funded by Research grant #201503 of the Forestry and Forest Products Research Institute. We also owe much to the useful advice of the anonymous reviewers.

Disclosure statement

No potential conflict of interest was reported by the authors.

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